

The Ambivalence of Relational Knowledge: Subject Intensity Asymmetry and the Structure of Well-Intentioned Erosion

Child Paper 3 — English Full Text v1.0 (Complete Academic Version)

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Abstract

Relational knowledge functions both as a condition for subject formation and as a medium through which subject-injury propagates. This paper formalizes a third pathway into subject-injury, distinct from rising conformity pressure and the institutional simplification of evaluative space: the asymmetry of subject intensity within a relational field.

We introduce *subject intensity* (S_r) as a relationally dependent composite index representing the degree to which a subject maintains difference-retention, recomputation capacity, and evaluative autonomy without fully externalizing its evaluative reference point to another party. The operative variable is not S_r itself but *subject intensity difference* (ΔS_r): the central question is not which subject is stronger, but which is structurally dependent on the other's frame.

When ΔS_r is large, the lower-intensity party is drawn toward a cost-minimizing path in which adopting the higher-intensity party's frame reduces the burden of maintaining its own difference-retention. This dynamic can arise even in relationships characterized by trust, care, and good intent, precisely because trust lowers the perceived cost of frame adoption. We term this *well-intentioned erosion*: subject-injury driven not by coercion, but by the relational outsourcing of recomputation through trust and cost minimization.

The boundary between temporary scaffolding and well-intentioned erosion is identified by three co-occurring signs: sustained decline in self-initiated judgment, unidirectional difference supply, and asymmetric suspension of evaluation. Isolated dependency constitutes scaffolding; the repeated co-occurrence of these three signs constitutes erosion. Trust is therefore neither inherently generative nor inherently erosive, but functions as an accelerant in either direction depending on whether bidirectionality of difference supply is structurally maintained.

A note on the theoretical frameworks

OMD (Organizational Metastability Dynamics), RD (Recomputation Dynamics), and RLAF (Reinforcement Learning from AI Feedback) are theoretical frameworks developed within the present research program and are introduced here as analytical constructs rather than references to previously published models under those names.

1. Introduction

1.1 The Problem

Why does the same relationship produce opposite results?

We often learn a great deal from people we trust. Yet at the same time, we can be drawn into their way of seeing things and lose our own evaluative orientation. Having someone who supports us makes it easier to maintain D (difference-retention) with a sense of security. Yet at the same time, we can find ourselves evaluating reality only through the supporter's frame.

Why does this reversal occur?

Existing theories of relational knowledge tend to portray the richness of knowledge through relationships and community-based connection in an affirmative light. Existing theories of domination treat relationships in terms of power, coercion, and suppression. Both offer important insights, but between them lies a domain that has not yet been described.

What occupies this domain is subject-injury without coercion, that is, subject-injury within relationships of good intent, trust, and support.

The question of this paper is therefore as follows: Why can a well-intentioned relationship become a medium for subject-injury? What is happening when it does? What structural conditions can prevent it?

The pathway described in Child Paper 2 — rising conformity pressure C — presupposes some form of external pressure. The pathway described in Child Paper 5 — the institutional simplification of evaluative space — presupposes institutional design. Yet in reality, subject-injury can proceed quietly even within relationships that lack external pressure or institutional simplification, indeed, within relationships that are rich and full of trust.

This paper formalizes that pathway as the asymmetry of subject intensity.

What matters is that the third pathway identified in this paper is independent in the following sense: the asymmetry internal to the relational field itself functions as the entry condition, rather than being a derivative of rising conformity pressure C or institutional simplification. The third pathway can operate even in environments where C is low and institutional simplification is absent.

What is at stake in this paper is not whether a relationship is good or bad, but how even a well-intentioned relationship can structurally reverse into erosion.

1.2 Organization of This Paper

Section 2 formalizes the ambivalence of relational knowledge and organizes the conditions under which the generative and erosive directions diverge. Section 3 defines subject intensity S_r and subject intensity difference ΔS_r , and describes the dynamics of the third pathway. Section 4 refines the conditions of divergence and presents operationalization candidates. Section 5 derives the conditions for relational protection in correspondence with the four principles of Child Paper 5. Section 6 organizes the limitations and directions for future work.

2. The Ambivalence of Relational Knowledge and the Conditions of Divergence

2.1 Defining Relational Knowledge

Building on Chapter 3 of the parent paper and the account in Child Paper 2, this paper defines relational knowledge as follows:

Relational knowledge is the process through which the interaction of multiple subjects gives rise to a form of recomputability that no individual subject could have generated alone, together with the knowledge structure sedimented in the relational field through that process.

The core of this definition lies in the condition “could not have been generated alone.” It is structurally difficult for an individual to betray its own existing frame by itself. Difference inputs capable of destabilizing a frame are supplied through interaction with others in the relational field. Relational knowledge is understood as the structure of that supply process.

2.2 The Three Conditions for the Generative Direction

When relational knowledge functions in a direction that enhances the recomputability of subjects, at least three conditions obtain.

Condition 1: Retention of difference. When $\text{Var}(\varphi)$ is maintained in the relational field — when minority views, alternative interpretations, and questions about premises are not invalidated — each subject continues to receive a supply of D.

Condition 2: Preservation of the history of difference. Difference is not treated as if it never existed; traces of difference remain in the relational field even after agreement is reached. This accumulation forms the depth of relational knowledge.

Condition 3: The emergence of a third subject. When a field of recomputation arises between two parties that cannot be reduced to either of them, a relational subject, that is, the third subject, emerges. This is a phase space for recomputation that extends beyond the individual.

2.3 The Three Pathways of the Erosive Direction

When the same relationship moves in an erosive direction, there are at least three independent entry points.

Pathway 1: Rising conformity pressure C (the primary pathway in Child Paper 2). The pathway in which dependence on external evaluative axes intensifies and the cost of maintaining difference rises. Explicit pressure is the leading factor.

Pathway 2: The institutional simplification of evaluative space (the second pathway in Child Paper 5). The pathway in which the design of the relational field itself invalidates difference. This can occur even when C is low. Institutional or environmental design is the leading factor.

Pathway 3: The asymmetry of subject intensity (the distinctive claim of this paper). The pathway in which, without explicit coercion or institutional simplification, the lower-intensity party voluntarily relinquishes D by virtue of the subject intensity difference ΔS_r in the relational field. This pathway can operate even within well-intentioned relationships.

The central claim of this chapter:

The three pathways are independent entry conditions; the third pathway can operate even in environments where C is low and institutional simplification is absent. Accordingly, the defense against subject-injury must go beyond lowering C and expanding evaluative space — it must include addressing ΔS_r in the relational field.

3. Subject Intensity Difference ΔS_r : Formalizing the Third Pathway

3.1 Defining Subject Intensity S_r

Subject intensity S_r is a relationally dependent composite index that summarizes the degree to which a subject can maintain D, M, and T_eval in a relatively independent way within a given relational field — without fully externalizing its final evaluative reference point to the other party's frame.

Three points are important in the definition of S_r .

First, S_r is not an intrinsic attribute of the subject but a **relational quantity**. The same subject may have high S_r in one relational field and low S_r in another. It must be read not as a lack of capacity but as a structural condition of the relational field.

Second, S_r is a composite of the degree to which D, M, and T_eval are maintained independently; it does not introduce a new foundational variable. It is continuous with the variable system of Child Paper 2. More precisely, S_r is not a simple sum but a quasi-multiplicative index: it summarizes how far the self-operated functioning of D, M, and T_eval in the relational field can be maintained without externalizing the final evaluative reference point, and it has a non-substitutable structure in which a sustained decline in any one of them can accelerate the externalization of the final evaluative reference point.

Third, the operative central variable of this paper is not S_r itself but **ΔS_r (subject intensity difference)**. The question is not “which subject is stronger?” but “which subject in the relational field is dependent on the other's frame?”

3.2 The Dynamics Produced by ΔS_r

What happens in a relational field where ΔS_r is large?

The higher-intensity party evaluates difference through its own frame and continues to supply D. The lower-intensity party can substantially reduce the cost of recomputation by evaluating reality through the higher-intensity party's frame.

This is rational as cost minimization. At the same time, as the lower-intensity party ceases to operate its own D, M, and T_eval, these capacities weaken. Circuits that are not used may persist at lower maintenance cost, but their function also atrophies.

This is the decisive difference from Child Paper 2. The depletion of E_r_eff described in Child Paper 2 is preceded by the exhaustion of resources. In the third pathway described here, the relational outsourcing of recomputation comes first. What is happening in the third pathway is not the cessation of recomputation as such, but the process in which the repeated relational outsourcing of recomputation causes the self-operated functioning of D, M, and T_eval to thin out.

3.3 The Mechanism by Which Well-Intentioned Erosion Arises

Why does this occur more readily in relationships of good intent, trust, and support?

In high-trust relationships, the cost for the lower-intensity party of adopting the higher-intensity party's frame declines. The judgment that "this person's way of seeing is trustworthy" opens a lower-cost path than maintaining D on one's own. In relationships of support, the higher-intensity party actively supplies difference and provides evaluative frameworks, which makes it easier for the lower-intensity party's self-operated recomputation to be treated as unnecessary.

It should be noted that the trust-cost account in this paper is not a formal model but a quasi-formal description of how ΔS_r makes the externalization of D lower-cost. The framework of reading trust as a discount on evaluative cost is positioned not as something to be developed into equations, but as an auxiliary conceptualization for intuitively describing the direction of the dynamic.

What matters is that this occurs without malicious intent. The higher-intensity party is trying to support. The lower-intensity party is trying to respond to trust. As a consequence of this interaction, the lower-intensity party's D is depleted.

The boundary between learning or support and erosion must be made clear here. The problem is not the act of referencing another's frame as such. What is problematic is that when this is repeated, the frequency with which the lower-intensity party operates its own difference-retention and evaluative circuit declines, and the bidirectionality of recomputation in the relational field is lost. A relationship in which one uses another's frame as temporary scaffolding while continuing to maintain one's own D is not erosion but part of a generative direction. A relationship in which the operational frequency of D sustains a persistent decline and bidirectionality ceases to be restored is what this paper describes as well-intentioned erosion.

The determination of well-intentioned erosion is made not by the act of referencing another's frame itself, but by the repeated co-occurrence of three signs: (1) sustained decline in self-initiated judgment, (2) unidirectional difference supply, and (3) asymmetric suspension of evaluation. A single instance of dependency constitutes scaffolding; when these three signs repeatedly co-occur, the progression is described as erosion.

Well-intentioned erosion is the process in which, not through malicious intent or coercion but through the interaction of trust and cost minimization, the recomputability of the lower-intensity party is continuously outsourced relationally.

3.4 Application to Four Contexts

This dynamic appears across at least four typical contexts.

Education (teacher-student, parent-child): The teacher or parent functions as the higher-intensity party; the student or child as the lower-intensity party. The trust that “what the teacher says is correct” can work in a direction that makes the student's own T_{eval} operation unnecessary.

Support and care: The supporter is the higher-intensity party; the person being supported is the lower-intensity party. The more attentive the support, the more readily it creates conditions in which the supported party does not need to use their own frame.

Collaboration with AI: When AI functions as the higher-intensity party. The more one depends on an AI that provides high-precision output, the greater the risk that the human side's T_{eval} ceases to operate. This is structurally analogous to the RLAF pathway in Child Paper 2, but what is distinctive here is that it can obtain even in well-intentioned AI design.

Mentoring and coaching: The mentor is the higher-intensity party; the mentee is the lower-intensity party. The higher the quality of mentoring, the more readily the mentor's frame rather than the mentee's becomes the standard.

4. Refining the Conditions of Divergence and Their Operationalization

4.1 A Refined Table of Generative and Erosive Directions

Condition	Generative direction	Erosive direction
$Var(\varphi)$	Maintained	Converging
Direction of D supply	Bidirectional and mutual	Unidirectional (higher-intensity $\rightarrow$ lower-intensity)
ΔS_r	Small and symmetric	Large and asymmetric
Mutual tolerance of suspension	Both parties' suspension is treated as valid	The lower-intensity party's suspension is invalidated
Record of frame updating	L3 has occurred bidirectionally	L3 occurs only for the higher-intensity party
Level of trust	High (facilitates generative)	High (also facilitates erosion)

Condition	Generative direction direction)	Erosive direction
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Note: The key feature of this table is that the level of trust operates in both directions — generative and erosive. The higher the trust, the more readily emergence occurs; and the more readily well-intentioned erosion occurs. This is the structural basis for why preventing well-intentioned erosion is difficult.

More precisely, **trust can function simultaneously as a condition for emergence and as an accelerant for erosion**. This ambivalence is the distinctive object of description in this paper — one that is absent from the pathways described in Child Papers 2 and 5.

A summary note on the ambivalence of trust

Trust is, in itself, neither inherently generative nor inherently erosive. Trust works in the generative direction when it supports bidirectional recomputation, and in the erosive direction when it makes unidirectional dependence lower-cost. This distinction is the basis for the paper's claim — not “break the trust relationship,” but “design the structure of bidirectionality.”

4.2 Operationalization Candidates

Direct measurement of S_r and ΔS_r is not feasible at the present stage. The following observable traces are proposed as proxy indicator candidates, however.

Bidirectionality of D supply: Is it the case in the relational field that instances in which A destabilizes B's frame and instances in which B destabilizes A's frame are symmetrically on record?

Mutual tolerance of suspension: Are both parties' expressions of “not yet decided” or “I want to reconsider” treated as equally valid? A relationship in which only one party can suspend indicates a high- ΔS_r state.

Record of frame updating: Has L3-level updating — the questioning of premises, the revision of evaluative axes — occurred bidirectionally within the relationship? L3 in only one direction is a signal of asymmetry.

Change in dependency rate: Looking across time, is self-initiated judgment by the lower-intensity party — judgment made without reference to the higher-intensity party's frame — declining?

These proxy indicators are preliminary and intended to support future falsifiable designs rather than to constitute measurement instruments in their current form.

5. The Conditions for Relational Protection

5.1 Why “Design” Rather Than “Protection”

Addressing the third pathway should move not in the direction of “protecting the lower-intensity party” but in the direction of “establishing the design conditions for a relational field that does not allow ΔS_r to expand.”

The reason has the same structure as the subject protection argument in Child Paper 5. Moral demands on individuals — “form your own views,” “do not become dependent” — do not function unless the structural conditions are in place. What is at stake is the design of the relationship.

5.2 The Four Conditions for Relational Protection

The core of relational protection is the maintenance of a bidirectional structure in which the lower-intensity party can reference the higher-intensity party’s frame while avoiding full identification with it, and can continue to maintain its own D. Translating the four principles of Child Paper 5 to the relational scale yields the following four conditions.

It should be noted that the four conditions in this paper are not a reduced copy of the four principles of Child Paper 5. They are positioned as a relational-scale translation of institutional principles into the local structure of bidirectionality maintenance in a two-party relationship. Child Paper 5 handles institutional design; this paper handles the response to asymmetry in the relational field. This division of labor makes the two papers complementary.

Condition 1 (relational version of two-tier evaluation): The relational field has a structure in which “achievement evaluated by the other’s evaluative axis” and “recomputation by one’s own evaluative axis” are independently recorded and recognized. A paradigmatic example is a teacher evaluating a student’s alternative framing of the problem on a dimension independent of whether it is a correct answer.

Condition 2 (mutual tolerance of suspension): Both parties’ suspension and reconsideration are treated as equally valid. It is necessary that the supporter’s “I’m not sure” or “let’s think about it together” functions as the practice of questioning rather than as a sign of insufficient knowledge.

Condition 3 (recording and preserving difference): The differences of the lower-intensity party — alternative views, questions about premises, minority opinions — are recorded so that they do not disappear from the relational field. Not simply acknowledged with “there was that view too,” but preserved in a form that remains traceable.

Condition 4 (tolerance of partial non-participation): The relationship does not require the lower-intensity party’s full identification with the higher-intensity party’s frame. The relationship protects the space in which the lower-intensity party can say “I want to think about this using a different evaluative axis from yours.”

5.3 The Structural Conditions for Reducing ΔS_r

The four conditions above also function to reduce ΔS_r itself. When the higher-intensity party relativizes its own frame and actively receives the difference supply from the lower-intensity party, the bidirectionality of D supply in the relational field is restored.

The goal here is not complete symmetrization. The aim is not to bring ΔS_r to zero, but to prevent the unidirectional relationship in which the expansion of ΔS_r connects to the sustained substitution of the lower-intensity party's recomputability. Maintaining bidirectionality is the goal; eliminating asymmetry is not.

6. Discussion, Limitations, and Future Directions

6.1 Contributions of This Paper

This paper makes three distinctive contributions.

First, it formalizes subject intensity asymmetry (ΔS_r) as the third pathway into subject-injury. This is an independent entry condition from rising conformity pressure and institutional simplification; it can operate even in environments where C is low and simplification is absent.

Second, it describes the structure of well-intentioned erosion. It shows the process by which subject-injury proceeds not through malicious intent or coercion but through the interaction of trust and cost minimization.

Third, it defines S_r as a relationally dependent composite index derived from the variable system of Child Paper 2 — D, M, and T_{eval} — while maintaining consistency with the overall framework and opening questions distinctive to the relational field.

6.2 Limitations

The unresolved problem of operationalization: The measurement problem for S_r and ΔS_r remains. The proxy indicator candidates proposed in this paper require future measurement designs capable of rigorously distinguishing structural conditions from apparent lack of capacity.

The distinction between well-intentioned erosion and educational guidance: This paper does not identify every asymmetric relationship with subject-injury. Some degree of asymmetry is inevitable in education, support, and care, and is not in itself problematic. What becomes problematic is when asymmetry transitions from functioning as temporary scaffolding to functioning as a structure that continuously substitutes for the lower-intensity party's recomputability. The presence or absence of this transition is the boundary that separates generative support from well-intentioned erosion.

Scale limitation: This paper has analyzed primarily dyadic relationships between individuals. Application to relational fields of three or more parties, and to organizational or institutional scales, is deferred to future work.

The problem of causal direction: Whether the expansion of ΔS_r causes erosion, or whether the process of erosion expands ΔS_r , is difficult to determine in terms of temporal order at the present stage. Longitudinal research is required.

A testable implication of this framework is that relationships marked by persistently large ΔS_r will exhibit a time-series decline in self-initiated judgment on the lower-intensity side, even in the absence of explicit coercion. Future work should formulate at least one longitudinal falsifiable hypothesis of this form and develop the observational protocol required to test it.

6.3 Connection to Child Paper 1

What this paper has made clear is that a precise description of S_r requires precision in the definition of the subject itself. What does it mean to be in a state capable of recomputation through one's own frame? How does this correspond to the difference between quasi-subject and subject? — these are questions that Child Paper 1 (*Subject and Quasi-Subject Redefined*) is positioned to answer.

If Child Paper 1 provides the foundational definitions, the definition of S_r in this paper can be described with greater precision. Conversely, the perspective of “ S_r as relational quantity” developed in this paper may contribute a relationally dependent dimension to the subject definition of Child Paper 1.

6.4 Civilizational Scope

The broadest implication of the structure of well-intentioned erosion is as follows.

The more advanced a society becomes, the more people function through the division of labor, specialization, and networks of trust. This is the foundation of productivity and collaboration. At the same time, however, division of labor and specialization structurally produce ΔS_r . The more domains in which self-initiated recomputation is no longer required, the more D , M , and T_{eval} in those domains can weaken.

This can be read as a civilizational-scale Region B: the process in which individual capacities are maintained while the recomputability of specific domains collectively declines.

This question is the scope opened by Chapter 7 of the parent paper, and the endpoint of this paper is the point at which it connects to that scope.

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interlocutors across successive drafts for conceptual testing, reformulation, and critical comparison. Final decisions regarding theory, framing, interpretation, wording, and all claims presented in this manuscript are the sole responsibility of the author.

References

Berger, P. L., & Luckmann, T. (1966). *The social construction of reality: A treatise in the sociology of knowledge*. Anchor Books.

Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. Plenum Press.

Emirbayer, M., & Mische, A. (1998). What is agency? *American Journal of Sociology*, 103(4), 962–1023. <https://doi.org/10.1086/231294>

Putnam, R. D. (2000). *Bowling alone: The collapse and revival of American community*. Simon & Schuster.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78. <https://doi.org/10.1037/0003-066X.55.1.68>

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.

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